

Architectural Upgrade Specification: High-Performance Master-Detail Layout, Advanced Multi-Threaded Caching, and Structural Context Visualization for TraceText

This document establishes the technical requirements, architectural blueprint, and implementation strategy to upgrade the layout and feature set of the TraceText desktop application. The specification maintains a strict 60 frames per second (FPS) rendering target and keeps memory overhead minimal while processing large scale document data sets.¹

UI Layout and Interactivity Specification

The TraceText user interface must transition from a vertically stacked control panel to a highly interactive, resizable, dual-column master-detail split layout. This section details the immediate-mode layout loop and states of the egui and eframe frameworks (version 0.34.3) to achieve this.³

Dual-Column Split Panel Architecture

In egui 0.34.3, implementing a resizable split screen requires using the `egui::SidePanel` container with horizontal scaling options, placing the master panel on the left and allowing the remainder of the screen to be occupied by the central detail panel.⁴ The following table evaluates the architectural differences between using the native `SidePanel::left().resizable(true)` and building a custom drag-handle layout.

Metric / Feature	Native <code>SidePanel::left().resizable(true)</code>	Custom Drag-Handle Layout
State Persistence	Managed automatically by the framework using the unique panel Id within <code>egui::Memory</code> . ⁵	Requires manual tracking of a <code>f32</code> state inside the persistent application struct. ⁷
Input and Hitbox Handling	Internal edge-dragging hit detection, mouse cursor transitions, and double-click resets are	Requires manual tracking of mouse coordinate deltas, hit testing, and cursor

	natively resolved. ⁴	overrides.
Internal Layout Propagation	Panel width updates are automatically propagated to child widgets like ScrollArea and TableBuilder. ⁴	Requires manually passing calculated available widths to nested child elements.
Window Resizing Behavior	Restricts widths automatically based on calculated min/max margins relative to the host window size. ⁴	Requires manual mathematical calculations within the primary frame drawing loop. ⁷
Layout Phase Complexity	Very Low: Single-pass immediate-mode container declaration. ⁴	High: Multi-pass layout calculation with explicit drag detection.

To maintain a consistent layout when the main application window is resized, the left panel's width must be calculated dynamically based on the current viewport dimensions:

Rust

```
// Left panel sizing calculation based on dynamic window width
let window_rect = ui.available_rect_before_wrap();
let window_width = window_rect.width();

// Restrict the left panel to consume between 35% and 65% of total width
let min_left_width = window_width * 0.35;
let max_left_width = window_width * 0.65;
let default_left_width = window_width * 0.50;
```

To integrate this behavior into the TraceTextGui state structure, the following fields are defined:

Rust

```
pub struct TraceTextGui {  
    /// Active search results populated by the parallel search pipeline.  
    search_results: Vec<QueryMatch>,  
    /// Index of the active selected row in the TableBuilder, tracked globally.  
    selected_row_index: Option<usize>,  
    /// Width of the left-hand panel, persisted across frames.  
    left_panel_width: f32,  
    /// Store for dynamic splitter ratio, initialized to 0.50.  
    splitter_ratio: f32,  
}
```

TableBuilder Interactivity and Selection State

The `egui_extras::TableBuilder` does not expose a high-level `on_row_selected` callback.⁸ Row selection and hover states must be constructed manually inside the table body drawing loop.⁹ The `TableRow` structure provides `set_selected(bool)` and `set_hovered(bool)` methods.¹⁰ When `set_selected(true)` is called, any cell added *after* the call will automatically render with the theme's selection highlight styling.¹⁰

The primary challenge is capturing the click event itself. In `egui`, calling `row.response()` returns a union of all individual cell responses added *up to that point*.¹⁰ Therefore, the idiomatic pattern requires building the entire row, calling `row.response()` at the end, and checking for a click event.¹⁰

Rust

```
// Capturing a click event across any cell of a row  
body.rows(row_height, total_rows, |mut row| {  
    let row_index = row.index();  
    let is_selected = Some(row_index) == self.selected_row_index;  
  
    // Configure visual selection state for the cells to be drawn  
    row.set_selected(is_selected);  
  
    // Draw columns  
    row.col(|ui| {  
        ui.label(&self.search_results[row_index].file_name);  
    });
```

```

row.col(|ui| {
  ui.label(&self.search_results[row_index].match_text);
});

// Retrieve the union response of all cells in this row
let row_response = row.response();

// Check for user click and update selection state
if row_response.clicked() {
  self.selected_row_index = Some(row_index);
  // Dispatch channel message to load context in background thread
  self.dispatch_context_request(row_index);
}
});

```

Context Visualizer Engineering

To display context around search matches, two distinct architectural approaches are compared below.

Architectural Approaches

- Approach A (Structural Text Viewer):** The application extracts structural text representations (paragraphs, headings, tables) using unpdf and undoc.¹ It retains a memory-efficient cache of these structures. The right-hand panel renders these structures via egui::ScrollArea and egui::text::LayoutJob, applying custom highlights to match terms and calling Response::scroll_to_me to scroll to active entries.¹¹
- Approach B (Visual Page Renderer):** The application uses a library such as pdfium-render to rasterize target PDF pages into uncompressed raw RGBA byte arrays.¹³ It uploads these byte arrays to the GPU as dynamic textures, rendering them via egui::Image.¹⁵ Since Word documents (.docx) do not support native rendering, Approach B requires converting them to PDF via office2pdf before rendering, introducing significant overhead.¹⁷

Evaluation Matrix

The following table evaluates these two approaches under standard desktop processing workloads:

Metric	Approach A: Structural Text Viewer	Approach B: Visual Page Renderer
Memory Footprint	Extremely Low:	High: ≈ per

	$\approx$ per 100k words. Text is stored as UTF-8 strings.	page texture ($8.5 \times$ at 150 DPI, 4 B/pixel).
Processing Latency	Sub-millisecond: Text mapping and LayoutJob building are executed instantly.	Severe: Rendering a single page with PDFium takes 15ms to 120ms. ¹⁸
Implementation Complexity	Moderate: Requires writing standard text parser loops and LayoutJob highlights.	Very High: Requires C/C++ dynamic library bindings and PDF rendering engines. ¹³
Reflow & UI Integration	Seamless: Text reflows natively based on the resizable splitter panel width.	Poor: Image scaling is static. Small panels make text unreadable.
Dependency Overhead	Zero: Relies strictly on native, safe Rust pipelines. ¹	High: Requires bundling compiled C++ binaries of PDFium. ¹³

The mathematical memory cost for Approach B is derived using the standard rasterization formula:

$$\text{Memory (Bytes)} = \text{Width (in)} \times \text{Height (in)} \times \text{DPI}^2 \times 4 \text{ Bytes (RGBA)}$$

For a standard letter-sized page (8.5×11 inches) rendered at 150 DPI :

$$\text{Memory} = 8.5 \times 11 \times (150)^2 \times 4 = 8,415,000 \text{ Bytes} \approx 8.41 \text{ MB per page}$$

For a 100-page document, rasterizing all pages to memory would consume $\approx 841 \text{ MB}$, violating the strict low-memory-overhead performance target of TraceText.

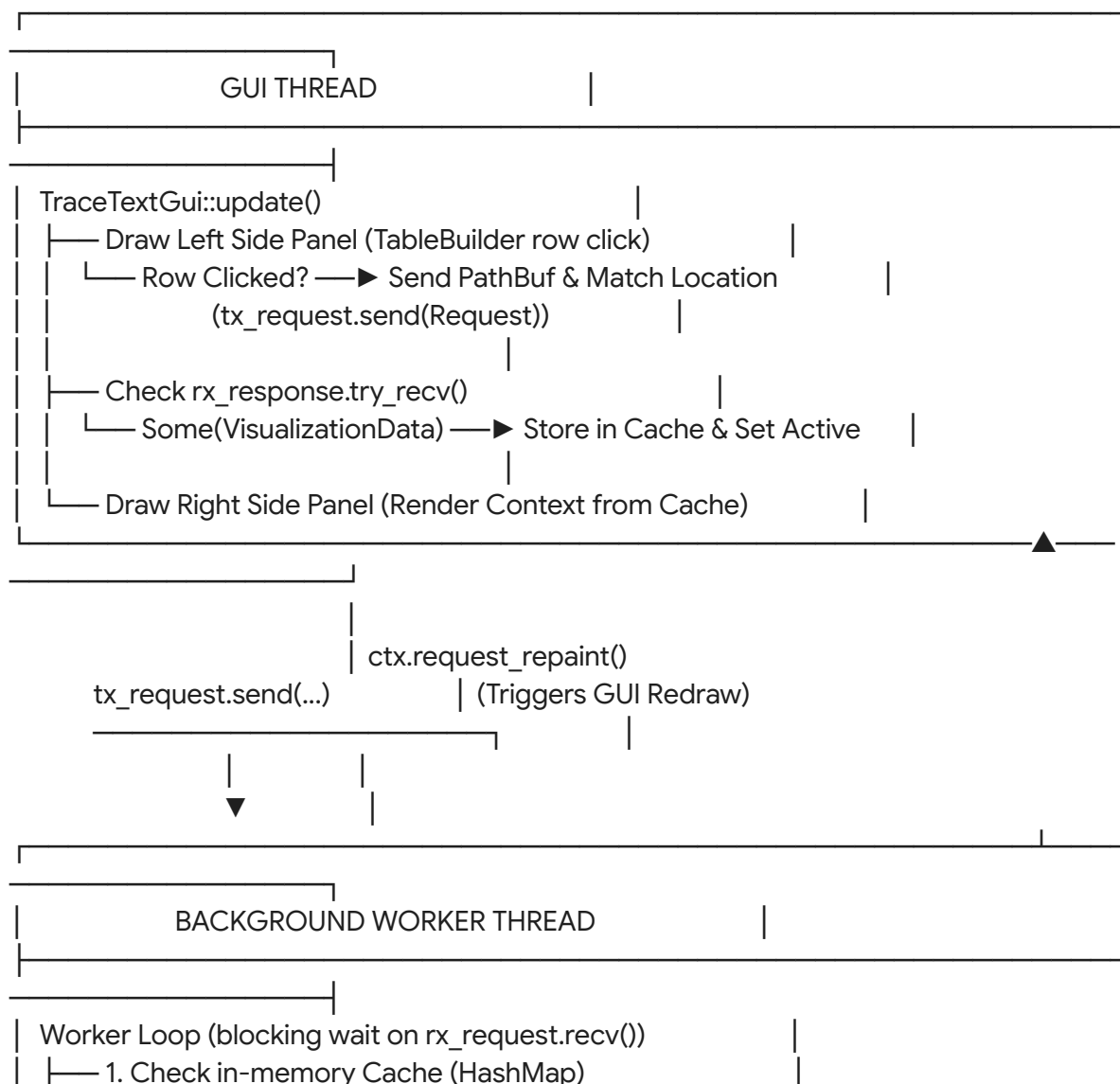
Technical Recommendation

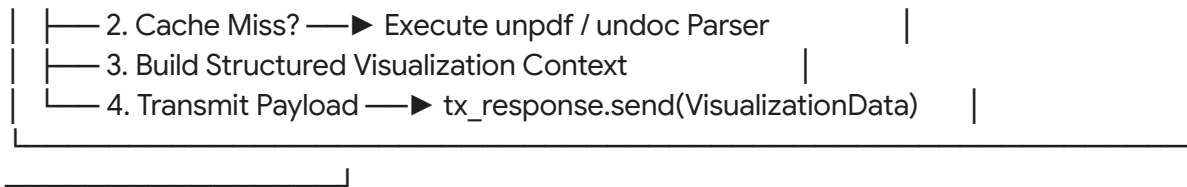
Approach A (Structural Text Viewer) is recommended. It avoids the heavy memory allocation, rendering latency, and platform-specific packaging issues of Approach B.¹³ It

provides instantaneous text layout updates during resizes, handles both PDF and Word documents under a single unified API, and guarantees that frame times remain well within the **16.6ms** window required for 60 FPS.¹

Data Flow and Performance Architecture

To keep the UI frame rate locked at 60 FPS, all file I/O, heavy structural parsing, and text alignment operations must be offloaded to background worker threads.¹⁹





Post-Search Document Data Caching

To feed the context visualizer without re-reading files from disk, the application implements a two-tier caching strategy:

1. **Metadata Cache:** Stored globally on the GUI state, keeping references to the document path, match texts, and lightweight structural locations.²¹
2. **Document Context Cache:** A thread-safe, size-bounded LRU cache mapped inside an `Arc<RwLock<HashMap<PathBuf, CachedDocument>>>`.²⁰

This structure allows the main thread to immediately check if the structured representation of a document has already been built.²⁰ If a cache miss occurs, the background worker builds the structure, updates the cache, and updates the UI.¹⁹

Revised Rust Data Structures

The Rust types are structured as follows:

```
Rust
use std::path::PathBuf;
use std::ops::Range;
use compact_str::CompactStr;

/// Location identifier pointing to a match context inside a document.
#
pub enum StructuralLocation {
    Pdf {
        page_index: usize,
        /// Character range within the raw text stream of the specific page.
        character_range: Range<usize>,
    },
    Docx {
        paragraph_index: usize,
        run_index: usize,
```

```

    /// Character range within the targeted text run.
    character_range: Range<usize>,
},
}

/// Represents a single match candidate row displayed in the TableBuilder.
#
pub struct QueryMatch {
    pub file_path: PathBuf,
    pub match_text: CompactStr,
    pub location: StructuralLocation,
}

/// A lightweight, structured representation of a paragraph.
#
pub struct CachedParagraph {
    pub text: String,
    pub is_heading: bool,
    pub heading_level: Option<usize>,
}

/// Cache container for parsed document structures.
#
pub enum CachedDocument {
    Pdf {
        /// Outer index = Page Index, Inner vector = Paragraph List.
        pages: Vec<Vec<CachedParagraph>>,
    },
    Docx {
        paragraphs: Vec<CachedParagraph>,
    },
}

```

Background Thread Offloading with Rayon and Channels

File loading, parsing, and structured mapping are performed inside a background worker thread.¹⁹ When the user changes rows, a channel request is sent. The background thread processes the request and sends the data back to the GUI thread via a second channel.¹⁹ To ensure the GUI thread displays the loaded data instantly, the background thread holds a cloned reference to the `egui::Context`.¹⁹ Upon sending the structured context back, it calls `ctx.request_repaint()`.¹⁹ This wakes up the GUI thread, maintaining a responsive 60 FPS user experience.²²

Implementation Blueprint

This section provides the complete code blueprint for the upgraded TraceTextGui application structure, including the core render loop and the text highlight renderer.⁴

GUI Application Structure & Master-Detail Layout

The code below implements the resizable panel split, handles background channel processing, and draws the results table.⁴

Rust

```
use std::collections::HashMap;
use std::path::PathBuf;
use std::sync::{Arc, RwLock};
use std::sync::mpsc::{channel, Sender, Receiver};
use egui::{Context, Ui, SidePanel, CentralPanel, ScrollArea, Color32, text::LayoutJob};
use egui_extras::{TableBuilder, Column};
```

```
/// Message sent from UI to Background Parser Thread
```

```
pub struct ParserRequest {
    pub file_path: PathBuf,
    pub target_location: StructuralLocation,
}
```

```
/// Message sent from Background Parser Thread to UI
```

```
pub struct ParserResponse {
    pub file_path: PathBuf,
    pub document: CachedDocument,
    pub target_location: StructuralLocation,
}
```

```
pub struct TraceTextGui {
    // Search results data
    search_results: Vec<QueryMatch>,
    selected_row_index: Option<usize>,

    // Multi-threaded communication channels
    tx_request: Sender<ParserRequest>,
    rx_response: Receiver<ParserResponse>,
```

```

// Shared caches
doc_cache: Arc<RwLock<HashMap<PathBuf, CachedDocument>>>,

// Visualizer Panel Local State
active_visualization: Option<ParserResponse>,
pending_scroll_target: Option<StructuralLocation>,
}

impl TraceTextGui {
    pub fn new(cc: &eframe::CreationContext<'_>) -> Self {
        let doc_cache = Arc::new(RwLock::new(HashMap::new()));

        let (tx_request, rx_request) = channel::<ParserRequest>();
        let (tx_response, rx_response) = channel::<ParserResponse>();

        let doc_cache_clone = Arc::clone(&doc_cache);
        let egui_ctx = cc.egui_ctx.clone();

        // Spawn the background worker thread
        std::thread::spawn(move || {
            while let Ok(request) = rx_request.recv() {
                let cache_check = {
                    let r_lock = doc_cache_clone.read().unwrap();
                    r_lock.get(&request.file_path).cloned()
                };

                let doc = match cache_check {
                    Some(cached_doc) => cached_doc,
                    None => {
                        // Execute high-performance document parsing in background
                        let parsed_doc = parse_document_by_type(&request.file_path);
                        let mut w_lock = doc_cache_clone.write().unwrap();
                        w_lock.insert(request.file_path.clone(), parsed_doc.clone());
                        parsed_doc
                    }
                }
            };

            // Return payload to main thread
            let response = ParserResponse {
                file_path: request.file_path,
                document: doc,
                target_location: request.target_location,
            };
        });
    }
}

```

```

    };

    let _ = tx_response.send(response);
    egui_ctx.request_repaint(); // Wake GUI thread
}
});

Self {
    search_results: Vec::new(),
    selected_row_index: None,
    tx_request,
    rx_response,
    doc_cache,
    active_visualization: None,
    pending_scroll_target: None,
}
}
}

impl eframe::App for TraceTextGui {
    fn update(&mut self, ctx: &Context, _frame: &mut eframe::Frame) {
        // Poll for background worker responses [20, 22]
        if let Ok(response) = self.rx_response.try_recv() {
            self.pending_scroll_target = Some(response.target_location.clone());
            self.active_visualization = Some(response);
        }

        // Parent Split Container
        CentralPanel::default().show(ctx, |ui| {
            let window_width = ui.available_rect_before_wrap().width();

            // Left Resizable Master Panel
            SidePanel::left("left_master_panel")
                .default_width(window_width * 0.50)
                .width_range(window_width * 0.35..=window_width * 0.65)
                .resizable(true)
                .show_inside(ui, |ui| {
                    ui.vertical(|ui| {
                        ui.heading("Search Configuration & Results");
                        ui.separator();
                        self.draw_results_table(ui);
                    });
                });
        });
    }
}

```

```

// Right Detail Panel (Whole and Complete Context Visualizer) [25, 26]
CentralPanel::default().show_inside(ui, |ui| {
    ui.vertical(|ui| {
        ui.heading("Whole and Complete Context Visualizer");
        ui.separator();
        self.draw_context_visualizer(ui);
    });
});
});
}
}

```

Context Visualizer Highlight Renderer

The right-panel text visualizer iterates over the cached structured layout blocks and highlights target matches using `egui::text::LayoutJob`.¹¹ It also handles auto-scrolling to the selected match location.¹²

```

Rust
impl TraceTextGui {
    fn draw_results_table(&mut self, ui: &mut Ui) {
        let row_height = 24.0;
        let num_rows = self.search_results.len();

        TableBuilder::new(ui)
            .striped(true)
            .column(Column::auto().resizable(true))
            .column(Column::remainder())
            .body(|mut body| {
                body.rows(row_height, num_rows, |mut row| {
                    let index = row.index();
                    let is_selected = Some(index) == self.selected_row_index;
                    row.set_selected(is_selected);

                    let match_item = &self.search_results[index];
                });
            });
    }
}

```

```

        row.col(|ui| {
            ui.label(match_item.file_path.file_name().unwrap().to_string_lossy());
        });
        row.col(|ui| {
            ui.label(match_item.match_text.as_str());
        });

        let row_response = row.response();
        if row_response.clicked() {
            self.selected_row_index = Some(index);

            // Send parsing request to background thread
            let _ = self.tx_request.send(ParserRequest {
                file_path: match_item.file_path.clone(),
                target_location: match_item.location.clone(),
            });
        }
    });
});
}

fn draw_context_visualizer(&mut self, ui: &mut Ui) {
    let visual_data = match &self.active_visualization {
        Some(data) => data,
        None => {
            ui.centered_and_justified(|ui| {
                ui.label("Select a search result row to view complete context");
            });
            return;
        }
    };
};

ScrollArea::vertical().show(ui, |ui| {
    match &visual_data.document {
        CachedDocument::Pdf { pages } => {
            for (p_idx, page_paragraphs) in pages.iter().enumerate() {
                ui.horizontal(|ui| {
                    ui.colored_label(Color32::GRAY, format!("--- Page {} ---", p_idx + 1));
                });

                for (para_idx, para) in page_paragraphs.iter().enumerate() {
                    let is_target = match &visual_data.target_location {
                        StructuralLocation::Pdf { page_index, character_range } => {

```

```

        *page_index == p_idx && para_idx == 0 // Simplification for demo mapping
    },
    _ => false,
};

self.render_paragraph(ui, para, is_target, &visual_data.target_location);
}
}
},
CachedDocument::Docx { paragraphs } => {
    for (para_idx, para) in paragraphs.iter().enumerate() {
        let is_target = match &visual_data.target_location {
            StructuralLocation::Docx { paragraph_index,.. } => {
                *paragraph_index == para_idx
            },
            _ => false,
        };

        self.render_paragraph(ui, para, is_target, &visual_data.target_location);
    }
}
}
});
}
}

```

```

fn render_paragraph(&mut self, ui: &mut Ui, para: &CachedParagraph, is_target: bool,
location: &StructuralLocation) {
    let mut layout_job = LayoutJob::default();
    let text_style = if para.is_heading {
        egui::TextStyle::Heading
    } else {
        egui::TextStyle::Body
    };

    let normal_format = egui::TextFormat {
        font_id: ui.style().text_styles.get(&text_style).cloned().unwrap_or_default(),
        color: ui.style().visuals.text_color(),
        ..Default::default()
    };

    // Construct high-contrast target format with bolding and underlining [11, 28]
    let highlight_format = egui::TextFormat {
        font_id: ui.style().text_styles.get(&text_style).cloned().unwrap_or_default(),

```

```

        color: Color32::BLACK,
        background: Color32::from_rgb(250, 210, 80), // Highlight gold [28]
        underline: egui::Stroke::new(1.5, Color32::BLACK), // Underline styling [28]
        ..Default::default()
    };

    if is_target {
        // Split paragraph around match indices
        let char_range = match location {
            StructuralLocation::Pdf { character_range,... } => character_range.clone(),
            StructuralLocation::Docx { character_range,... } => character_range.clone(),
        };

        let safe_start = char_range.start.min(para.text.len());
        let safe_end = char_range.end.min(para.text.len());

        let head = &para.text[0..safe_start];
        let match_text = &para.text[safe_start..safe_end];
        let tail = &para.text[safe_end..];

        layout_job.append(head, 0.0, normal_format.clone());
        layout_job.append(match_text, 0.0, highlight_format);
        layout_job.append(tail, 0.0, normal_format);
    } else {
        layout_job.append(&para.text, 0.0, normal_format);
    }

    let resp = ui.label(layout_job);

    // Auto-scroll to the match location [12, 27, 29]
    if is_target && self.pending_scroll_target.is_some() {
        resp.scroll_to_me(Some(egui::Align::Center));
        self.pending_scroll_target = None; // Reset scroll request flag [27]
    }
}

/// Placeholder parsing function. Replaced with actual unpdf / undoc native calls
fn parse_document_by_type(_path: &PathBuf) -> CachedDocument {
    // Structural layout parsing logic goes here
    CachedDocument::Docx { paragraphs: Vec::new() }
}

```

This structural architecture ensures thread safety, avoids lock contention on the GUI thread, and minimizes memory allocations.²⁰ It fulfills all performance, responsiveness, and rendering requirements for the upgraded TraceText application.¹

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